

YASSIN AYMAN

MECHATRONICS ENGINEERING STUDENT

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SUMMARY

Aspiring Mechatronics Engineer with a robust foundation in mechanical design, industrial automation, and rapid prototyping. Technical expertise spans the full engineering lifecycle—from precision CAD modeling and simulation to physical fabrication, advanced motor diagnostics, and process instrumentation. Experienced in MEP systems through professional internships and high-performance vehicle engineering. Eager to contribute to innovative engineering projects requiring rigorous technical precision and systematic problem-solving.

EDUCATION

Harvard Egypt School

High School Diploma Graduated: 2024

Suez Canal University

Bachelor of Science in Mechatronics Engineering | (GPA : 3.2) | Expected Graduation: 2028

EXPERIENCE

Mechanical Engineering Intern (MEP & HVAC) | GAFI (General Authority for Investment)

(Aug 2025 – Sep 2025)

Executed technical analysis of complex Mechanical, Electrical, and Plumbing (MEP) systems to ensure operational efficiency for large-scale facilities.

Managed troubleshooting and maintenance workflows for building automation systems, specifically targeting HVAC networks and elevator control boards.

Industrial Automation Trainee | ABB Bright Minds Summer Program

(Aug 2025 – Aug 2025)

Completed a rigorous two-stage automation curriculum consisting of 60 comprehensive online courses followed by 8 interactive engineering webinars.

Gained deep technical insights into industrial-scale power distribution, low/medium voltage systems, motor starting logic, and EV charging infrastructure.

TECHNICAL CORE

Mechanical & CAD: SolidWorks (CSWP, Simulation, Electrical, Additive Manufacturing, MBD), Kinematics, Precision Alignment, Rotor Balancing.

Industrial Automation: PLC Programming (Siemens SIMATIC), Variable Frequency Drives (VFD), Process Instrumentation, Servo Motor Diagnostics, Encoders/Resolvers.

Digital Fabrication: CNC Machining, Laser Cutting, 3D Printing (HP Technology), Rapid Prototyping.

Electronics & Software: Proteus (Circuit Design/PCB), Arduino, C/C++, Python (NumPy, Pandas), MATLAB.

PROJECTS

Formula Student 2026 | Mechanical Design & Systems Integration

- Managed the chronological timeline for design concept specifications, contributing to the end-to-end development of a high-performance formula-style vehicle.
- Executed detailed CAD modeling, structural simulations, and stress analysis to optimize vehicle kinematics and material weight-to-rigidity ratios.

Robotic Manipulator Arm | Mechanics & Control Logic

- Designed a multi-DOF robotic arm utilizing SolidWorks, executing comprehensive degree-of-freedom analysis and applying inverse kinematics for trajectory planning.
- Implemented real-time control systems via embedded microcontrollers to ensure precise pick-and-place operational accuracy.

Autonomous Robosumo Combat Robot | Mechatronics Integration

- Engineered a high-torque chassis in SolidWorks, achieving 1:1 manufacturing accuracy between the digital model and the physical build.
- Integrated autonomous sensor arrays and motor controllers for intelligent target detection and real-time movement execution.

Precision Gearbox Prototype | CAD-to-Physical Fabrication

- Engineered high-efficiency gear ratios using SolidWorks simulation to ensure structural integrity and mechanical efficiency.
- Successfully transitioned digital models into high-accuracy physical prototypes, validating all assembly clearances and mechanical fits.

CERTIFICATIONS AND SESSIONS

- **Dassault Systèmes:** Certified SOLIDWORKS Professional (CSWP); Associate Certifications in Simulation, Electrical Design, and Additive Manufacturing.
- **Siemens:** SIMATIC Basics of Programmable Logic Controllers (PLC) - SITRAIN Certification.
- **Industrial Motor Control:** Variable Frequency Drives (VFD) Training. Mastered drive commissioning, parameter configuration, and speed/torque regulation for industrial AC motors.
- **EISAC Automation (Servo Motors):** Servo Motor Diagnostics & Repair Training. Gained hands-on expertise in KE testing via oscilloscope, absolute/incremental encoder signal analysis, precision shaft alignment, and dynamic rotor balancing.
- **Egyptian Engineers Syndicate :** Industrial Instrumentation Training. Acquired practical skills in configuring, calibrating, and interfacing industrial process measurement devices.
- **ITIDA / CREATIVA / KAAF:** Digital Fabrication Certificate (Mastery in CNC and rapid prototyping).
- **Microsoft / Sprints:** 40-hour AI & Machine Learning Summer Camp; Introduction to Generative AI Concepts Certification.
- **ITIDA Gigs:** 3-month freelance preparation program for delivering independent engineering solutions.